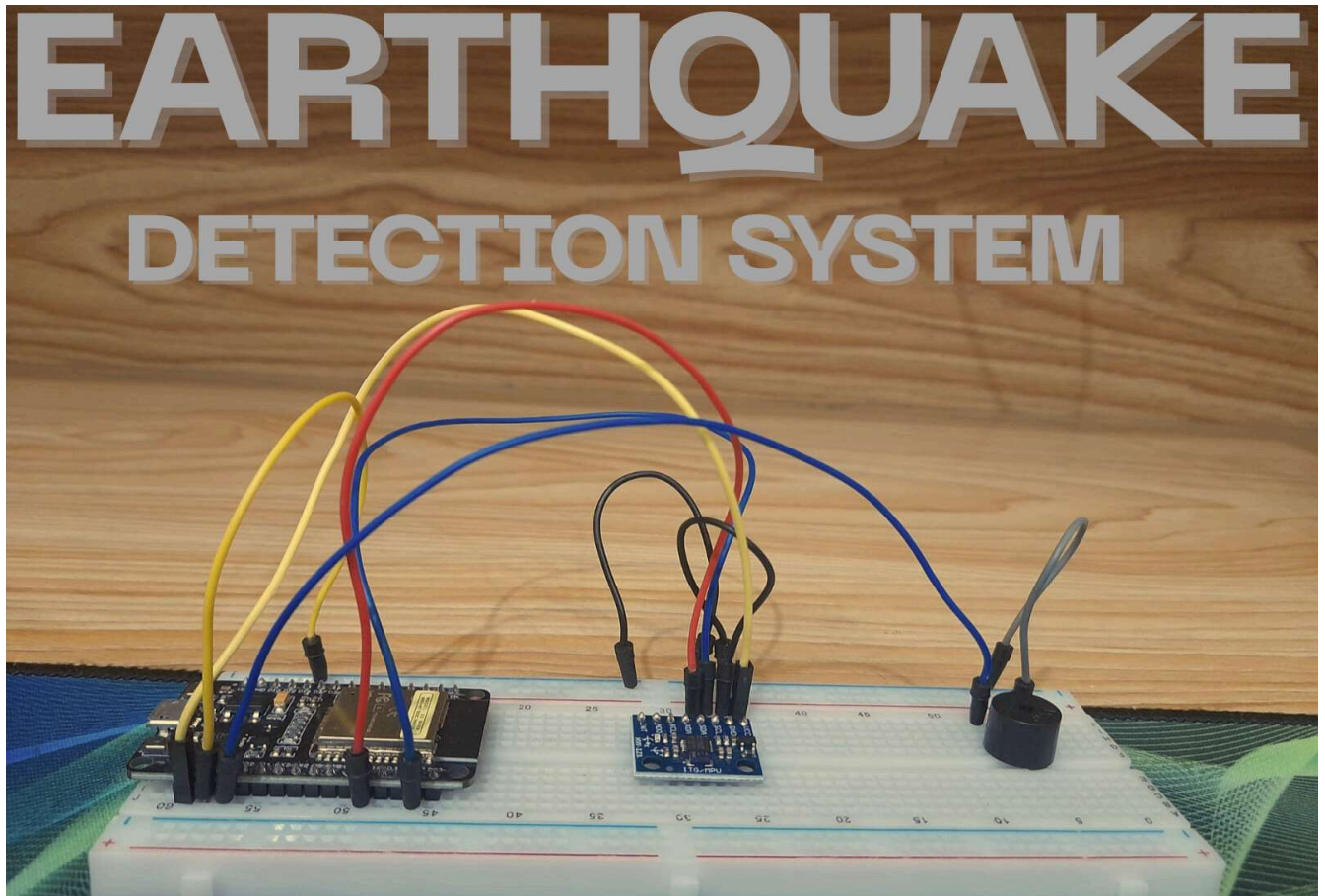




EARTHQUAKE DETECTION SYSTEM

System Overview

The Earthquake Detection System uses an ESP32 microcontroller and an MPU6050 accelerometer to continuously monitor vibrations.

When the detected G-Force crosses a threshold, the system:

1. Triggers a buzzer alert

2. Sends a Telegram notification with magnitude, time (GMT+6), and location (Dhaka, Bangladesh)

3. Displays real-time sensor data on a web dashboard

4. Stores live readings like G-force, axis values, and estimated Richter scale

The device connects to Wi-Fi and operates fully autonomously.

Setup and Operational Guide

1. Hardware Setup :

- ESP32 Dev Module
- MPU6050 Accelerometer
- Buzzer
- 5V Power Supply / USB Power
- Jumper Wires / Breadboard

2. Connection :

MPU6050 → ESP32 via I²C

- VCC → 3.3V
- GND → GND
- SDA → GPIO 21
- SCL → GPIO 22

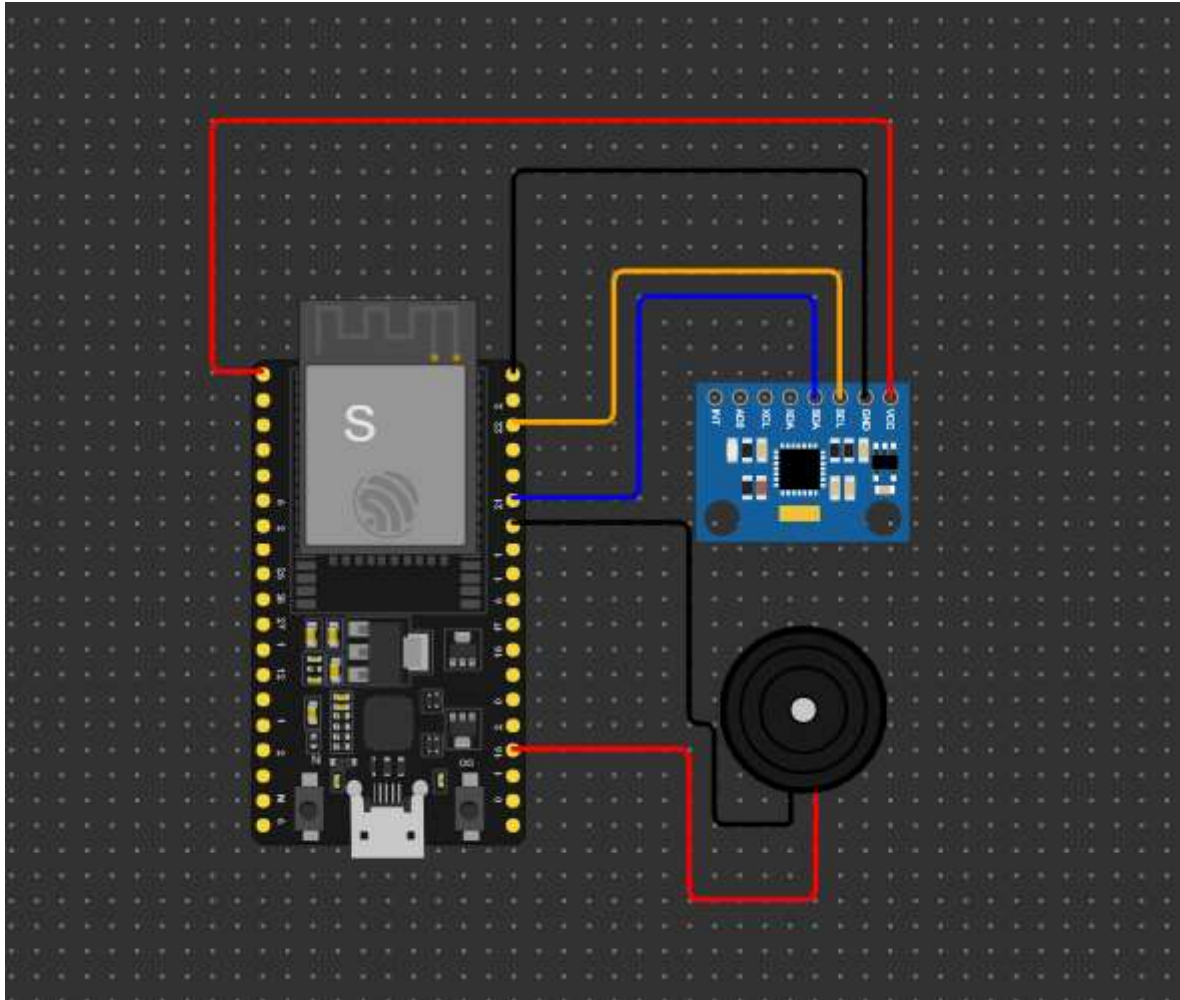
Buzzer → GPIO 15 (or any digital pin)

- Positive → GPIO
- Negative → GND

3. Install required libraries :

- Wire.h
- WiFi.h
- WebServer.h
- MPU6050.h
- UniversalTelegramBot.h

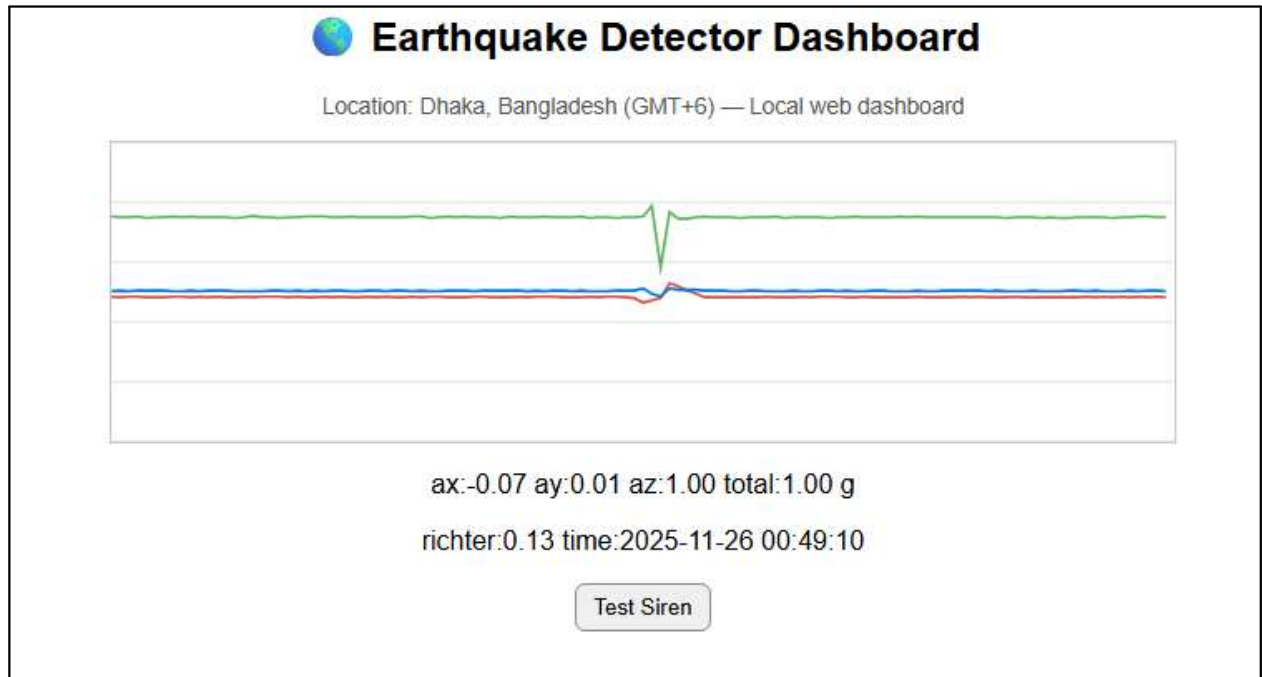
Circuit Setup



Code

[EARTHQUAKE.INO](https://github.com/robocorp/EARTHQUAKE.INO)

Dashboard



Telegram

 **EARTHQUAKE ALERT** ⚠️
bot

 **Earthquake Detected!**
Magnitude: 1.95
Richter
Time: 2025-11-25
01:23:07 (GMT+6)
Location: Dhaka, Bangladesh
G-Force: 1.88 g
nax:-0.06 ay:0.62 az:1.77

01:23 AM

How Does the System Work?

Motion Detection :

The MPU6050 reads X/Y/Z acceleration values hundreds of times per second.

G-Force Calculation :

$$\text{Total G} = \sqrt{ax^2 + ay^2 + az^2}$$

Richter Scale Estimation :

$$\text{Magnitude} = \log_{10}(\text{G-force}) \times \text{factor}$$

(custom formula tuned for vibration intensity)

Threshold Trigger :

If $G > \text{threshold}$ (example: 1.8g):

- Siren activates
- Telegram message sent with:
 - Magnitude
 - Time (Bangladesh GMT+6)
 - G-force
 - Ax, Ay, Az readings
 - Location: Dhaka, Bangladesh

Web Dashboard :

- Live graph of acceleration
- Real-time G-force
- System status (Safe / Alert)
- Test mode if MPU6050 is missing

Useful Tools

1. **Arduino IDE** – programming the ESP32
2. **Telegram BotFather** – create bot
3. **UniversalTelegramBot Library** – send alerts
4. **Real-time Serial Plotter** – debug acceleration
5. **Online Epoch Time Converter** – time debugging
6. **NTP Server** – for auto time sync

Summary

The Earthquake Detection System is a smart IoT solution built using ESP32, MPU6050, a web dashboard, and Telegram alerts.

It continuously monitors vibrations, calculates G-force, estimates magnitude, and instantly alerts users during abnormal movement.

The system is low-cost, fast, wireless, and ideal for:

- Smart homes
- Schools & offices
- Industrial monitoring
- Safety automation projects

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